

“Rewards and Consumption in the Credit Card Market” by Tianyu Han

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Credit Card Context

Many papers documenting behavioral mistakes in the credit card market

- Focus on costs of mistakenly incurring interest or late fees

(Beshears, Choi, Laibson, Madrian 18 Handbook Chapter)

- Increasing recent focus on rewards

(e.g., Mukharlyamov & Sarin, 24; Wang 24; Huynh, Nicholls and Shcherbakov 22; Agarwal, Presbitero, Silva, and Wix 23; Agarwal, Ang, Wang, and Zhang 24)

Data and Methods

Administrative data from large Chinese bank who offers credit cards.

RDD for platinum reward card eligibility to estimate effects of rewards on spending.

(“Fuzzy RDD is IV”, Angrist and Pischke, 2009)

Uses linked survey of 4,565 credit card users (response rate?).

(see Stantcheva 22 Ann. Review of Economics; Bergman, Chinco, Hartzmark, & Sussman 20)

Structural model considering implications for market structure & welfare.

Han's Hypothesis

1. Rewards → Spend More on Reward *and on Non-Reward* Categories

Reward spending \$64 (30%?), non-reward spending \$54 (5%?)

2. Why ↑ **Non-Reward Spending?** 'Complementary Ignorance'
(naïve consumers under-estimate how rewards change non-reward spending)

e.g., airline rewards → book flight.

But don't consider also spend on non-reward category (e.g. car rental)

3. Implications?

(i) Incentive banks to offer more generous rewards

(ii) Cross-subsidizes less to more sophisticated

(iii) **2.5% consumption cost per month**

My Comments Focus on Part 2

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Big Picture

- **Impressive Data Collection**
- **Convincing RDD design to show that rewards work**
- **Interesting new theory**

#1: Provide Detail on Platinum vs. Gold Rewards

Table 1. Example of Credit Card Rewards

	Gold	Platinum
5% off JD.com purchases	Y	Y
50% Starbucks/KFC	Y	Y
5% off gas/groceries	Y	Y
\$10 off movie tickets	Y	Y
Cashback on international flights		Y
Foreign airport pickup		Y
Travel insurance		Y
Hotel free buffet		Y
Travel medical insurance		Y

- Rewards (slightly?) reduce price of travel
- Are platinum rewards only for travel?
- Which rewards are % spending?
- Which are salient?
- (Beyond scope of this paper...
how do consumers value different rewards?)

Linking Effects to Literature

Good job of linking to Bursztyn, Ferman, Fiorin, Kanz, & Rao (17 QJE)

- platinum cards are status goods & ↑ in-person spending

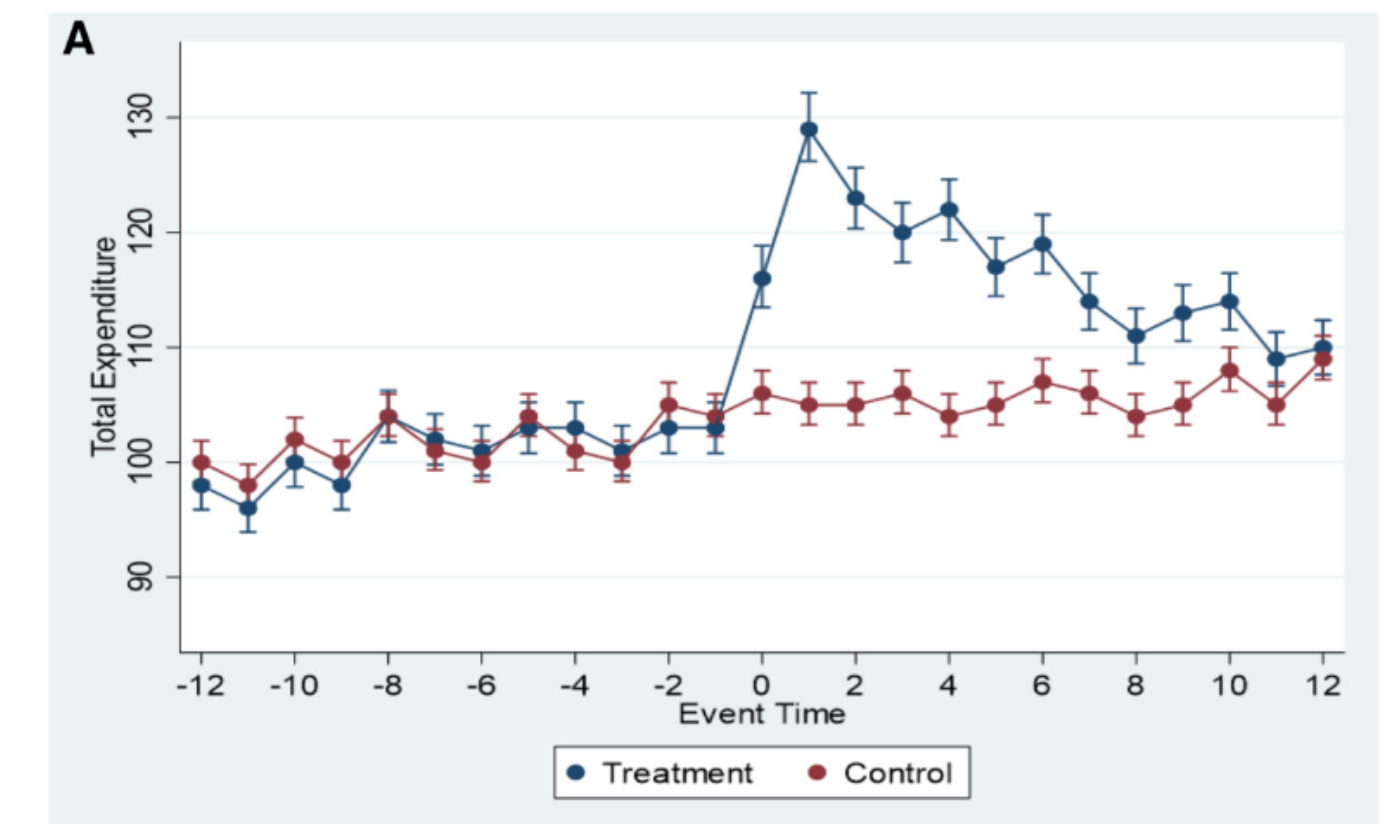
→ Han's paper shows effect on online transactions.



Gelman & Roussanov (24 RFS):

New credit card -> ↑ credit card spending

→ What is the marginal value of rewards?



Rewards ↑ spending is consistent with descriptive US evidence

(Agarwal, Presbitero, Silva, and Wix 23; Agarwal, Ang, Wang, and Zhang 24)

#2 Survey Measurement

Q6 is credit card rewards spending.

Q6 – Q4 is “non-rewards spending”

Q4. “What was your average monthly spending on non-durable in the past six months (excluding expenditure on durable goods such as housing, rent, and vehicle)?”

Q6. “For the consumption you have incurred over the past six months, on average, how much do you think are from the categories of goods that can earn rewards from your credit cards from XXX bank. For example, suppose your average monthly spending is 4,000 RMB. For 2,000 of the 4,000 RMB you have spent, you can earn cash back or enjoy a discount due to using your credit cards from XXX bank, then please enter 2,000.”

#2 Survey Measurement

Q6 is credit card rewards spending.

What is Q6 – Q4? A residual.

Q4. “What was your average monthly spending on non-durable in the past six months (excluding expenditure on durable goods such as housing, rent, and vehicle)?” **<-does *not* mention credit cards or bank XXX.**

**Q6-Q4 = Non-reward credit card spending + Non-credit card spending
+ other sources of survey error (e.g., rounding, not adding up)**

(see Sakaguchi, Gathergood, & Stewart 24; Herzog and Bartels 24)

Higher ‘error’ may have more unobserved spending and/or be less literate.

Suggestion: Decompose effects of rewards on non-reward credit card spending vs. non-credit card spending? Firms are incentivized to increase credit card vs. non-credit card spending as higher interchange.

#3 Timing

What I expected:

Elicit expectations -> quasi-random reward cards -> Observe future spending.

What this paper does:

Observe historical spending pre- and post-quasi-random reward card opening.

Survey understand of historical spending.

Don't think you can use backwards-looking responses to support paper's theory:

“Credit card rewards can lure naive consumers into booking reward-earning goods and services while overlooking the complementary purchases that will be necessary in the future.”

(see other works for how choose cards Agarwal, Chomsisengphet, Liu 10 JMCB; Agarwal, Chomsisengphet, Liu, Souleles 15 RCFS; Stango & Zinman, 16 RFS; Ru & Schoar, 23)

Not entirely clear why ‘suboptimal’ to spend more on non-reward consumption?

Cutting by endogenous variable.

Really we want to know the delta in expectations from gold vs. platinum

Consumers who who never had platinum cards

People who had platinum cards for 6+ months

People who had platinum cards for <6 months

#4 Policy Implications

Model shows welfare loss. What about counterfactual policies?

- Interchange caps of different levels (5.25% baseline)
- Simplify reward structure (e.g., can only be X% of any spending)
- (Iberti, Kulkarni, & Truffa, 23; US regulations do *not* standardize reward disclosures)
- More generally complexity in credit card rewards is area for researchers to explore

Han's Hypothesis

Rewards → Spend More on Reward *and on Non-Reward* Categories

CONVINCING

Why ↑ Non-Reward Spending? 'Complementary Ignorance'

APPEALING THEORY BUT I'M NOT CONVINCED ON EMPIRICS

Thanks!

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